

Deploying a Java-Based Application in Docker

This tutorial will help you create a simple application using Maven, install Docker on the Ubuntu operating system, create a JAR file using Jenkins, and finally, deploy a Java-based app.



Docker, an open platform, streamlines the building, testing, and deployment of applications, enhancing development efficiency by isolating applications from infrastructure or runtime environments. It provides UIs, CLIs, APIs and security features to package and run an application in lightweight containers. These containers have all the resources to run an application such as libraries, system tools, code, and runtime. In this article we will explore deploying a simple Java based application in Docker.

Creating a Java application using Maven

To build a package file for a sample application that we are going to run in a Docker container, Java and Maven need to be installed on the Ubuntu OS. Once that's done successfully, verify the versions using the 'java -version' and 'mvn -version' commands, respectively:

```
test@ubuntu:~/Desktop$ java -version
openjdk version "17.0.8.1" 2023-08-24
OpenJDK Runtime Environment (build 17.0.8.1+1-Ubuntu-0ubuntu122.04)
OpenJDK 64-Bit Server VM (build 17.0.8.1+1-Ubuntu-0ubuntu122.04, mixed mode, sharing)
```

```
test@ubuntu:~/Desktop$ mvn -version
Apache Maven 3.6.3
Maven home: /usr/share/maven
Java version: 17.0.8.1, vendor: Private Build, runtime: /usr/lib/jvm/java-17-openjdk-amd64
Default locale: en_IN, platform encoding: UTF-8
OS name: "linux", version: "6.2.0-37-generic", arch: "amd64", family: "unix"
```

Let's create a simple Java application using *mvn archetype:generate*:

```
test@ubuntu:~/Desktop$ mvn archetype:generate -DgroupId=com.osfy.app -DartifactId=my-app -DarchetypeArtifactId=maven-archetype-quickstart -DarchetypeVersion=1.4 -DinteractiveMode=false
```

Upon successful completion, you will see a success message like this:

```
[INFO] Project created from Archetype in dir: /home/test/Desktop/my-app
[INFO] -----
```